A self-contained spectral-shift proof package for the odd-cycle lower bound

High-density theorem, bipodal theorem, and the remaining Region-II target

Prepared for Hongseok Yang, Claude, and collaborators

July 4, 2026

Abstract

This note consolidates the spectral-shift work following the original file `paper.tex` and Claude's two-sided-shift note. It gives self-contained proofs of the algebraic reductions and records the cleaned conclusions. The main theorem proved here is the high-density case of the original conjecture: if a graphon W has edge density $p \geq 2/3$, then for every odd $m \geq 3$,

$$t(C_m, W) \geq p^m - p(1-p)^{m-1}.$$

The proof uses the two-sided spectral-shift identity, a complete-homogeneous-polynomial expansion, a mixture-of-diagonals reduction, a one-dimensional kernel representation, exact finite rational certificates for the short residual strip $m \leq 61$, and a new analytic argument for the infinite residual strip $m \geq 63$. The note also proves a separate all-density theorem for rank-two complement graphons, hence for all bipodal graphons. The full conjecture remains open only in the Region-II range $1/2 < p < 2/3$, where the remaining task is a one-frontier nonlinear spectral-payment lemma.

Contents

1 Problem statement and results

For a graphon $W : [0,1]^2 \to [0,1]$, write $t(H, W)$ for the homomorphism density of a finite graph H in W. Let

$$p = t(K_2, W), \qquad U = 1 - W, \qquad q = t(K_2, U) = 1 - p.$$

The original conjecture is the following.

Conjecture 1.1 (Odd-cycle lower bound)**.** *For every graphon W and every odd $m \ge 3$,*

$$t(C_m, W) \ge p^m - p(1-p)^{m-1} = p^m - pq^{m-1}.$$

The case $p \le 1/2$ is immediate, since the right-hand side is nonpositive. The results proved in this note are:

(i) Conjecture 1.1 holds for all graphons when $p \ge 2/3$.

(ii) Conjecture 1.1 holds for every graphon whose complement has the rank-two form

$$1 - W(x, y) = q + u(\psi(x) + \psi(y)) + \lambda\psi(x)\psi(y), \qquad \int \psi = 0, \quad \int \psi^2 = 1.$$

In particular, it holds for every bipodal graphon, with arbitrary block sizes.

The remaining range of the full conjecture is $1/2 < p < 2/3$.

2 Spectral setup

Let T_U be the integral operator of U on $L^2[0,1]$, and decompose $L^2 = \mathbb{R}\mathbf{1} \oplus \mathbf{1}^\perp$. Write

$$T_U\mathbf{1} = q\mathbf{1} + g, \qquad g \in \mathbf{1}^\perp, \qquad A = P_{\mathbf{1}^\perp} T_U P_{\mathbf{1}^\perp}.$$

Then, relative to $\mathbb{R}\mathbf{1} \oplus \mathbf{1}^\perp$,

$$T_U = \begin{pmatrix} q & g^* \\ g & A \end{pmatrix}.$$

Let μ denote the spectral measure of A weighted by g:

$$\mu(B) = \langle g, E_A(B)g \rangle, \qquad s_j := \int \lambda^j \, d\mu(\lambda) = \langle g, A^j g \rangle.$$

The earlier note `paper.tex` uses the standard compression bound $\sigma(A) \subseteq [-1/2, 1/2]$ and the trace-square budget

$$\mathrm{Tr}(A^2) + 2\|g\|_2^2 \le pq.$$

The arguments below only need the spectral support bound for the certificate domain.

Define formal power series

$$R_W(z) = \int \frac{d\mu(\lambda)}{1+\lambda z}, \qquad \mathcal{L}_W(z) = -\log\left(1 - \frac{z^2 R_W(z)}{1-pz}\right), \tag{1}$$

$$R_U(z) = \int \frac{d\mu(\lambda)}{1-\lambda z}, \qquad \mathcal{L}_U(z) = -\log\left(1 - \frac{z^2 R_U(z)}{1-qz}\right), \tag{2}$$

and for odd m set

$$S_m = m[z^m](\mathcal{L}_W(z) + \mathcal{L}_U(z)). \tag{3}$$

These are formal identities: $[z^m]$ depends only on q, p and the finitely many moments $s_0, \dots, s_{m-2}$.

3 The exact two-sided identity

Theorem 3.1 (Two-sided spectral-shift identity)**.** *For every graphon W and every odd $m \ge 3$,*

$$t(C_m, W) + t(C_m, U) = p^m + q^m + S_m. \tag{4}$$

Consequently Conjecture 1.1 is equivalent to

$$t(C_m, U) \le q^{m-1} + S_m. \tag{5}$$

Proof. First assume that U has finite rank. Since

$$T_W = J - T_U = \begin{pmatrix} p & -g^* \\ -g & -A \end{pmatrix},$$

conjugating by the sign flip on $\mathbf{1}^\perp$ shows that T_W is isospectral to

$$M = \begin{pmatrix} p & g^* \\ g & -A \end{pmatrix}.$$

Schur's determinant formula gives

$$\det(I - zM) = \det(I + zA)(1 - pz)\left(1 - \frac{z^2\langle g, (I + zA)^{-1} g\rangle}{1 - pz}\right),$$

$$\det(I - zT_U) = \det(I - zA)(1 - qz)\left(1 - \frac{z^2\langle g, (I - zA)^{-1} g\rangle}{1 - qz}\right).$$

Taking $-\log$ and using

$$-\log\det(I - zX) = \sum_{j\ge 1} \frac{\mathrm{Tr}(X^j)}{j} z^j,$$

we extract the coefficient of z^m and add the two identities. Since m is odd, the contributions from $\det(I + zA)$ and $\det(I - zA)$ cancel. The two scalar hub factors contribute $p^m + q^m$, and the two remaining factors contribute S_m. This proves (4) in finite rank.

For a general graphon, approximate U in L^2 by step kernels. Cycle densities are continuous under L^2 convergence for fixed cycle length, and S_m is a polynomial in the finitely many quantities $q, s_0, \ldots, s_{m-2}$, each of which is continuous under Hilbert–Schmidt convergence. Hence the identity passes to the limit.

Finally,

$$\begin{aligned} t(C_m, W) - (p^m - pq^{m-1}) &= q^m + pq^{m-1} + S_m - t(C_m, U) \\ &= q^{m-1} + S_m - t(C_m, U), \end{aligned}$$

which proves the equivalence. □

4 The path-shift defect and its exact expansion

Since $0 \le U \le 1$, deleting one edge from the odd cycle gives

$$t(C_m, U) \le t(P_{m-1}, U) =: x_{m-1}.$$

By Theorem 3.1, it is therefore enough to prove nonnegativity of the path-shift defect

$$\Phi_m := q^{m-1} + S_m - x_{m-1}. \tag{6}$$

The resolvent identity for the block matrix gives

$$\sum_{j \ge 0} x_j z^j = \langle \mathbf{1}, (I - zT_U)^{-1} \mathbf{1} \rangle = \frac{1}{1 - qz - z^2 R_U(z)}. \tag{7}$$

For $d \ge 0$, let h_d denote the complete homogeneous symmetric polynomial of degree d, with $h_0 = 1$.

Theorem 4.1 (Exact expansion of Φ_m). *Let $m \ge 3$ be odd. Then*

$$\Phi_m = \sum_{r=1}^{(m-1)/2} \int \cdots \int \mathcal{P}_{m,r}(q; \lambda_1, \ldots, \lambda_r) \prod_{i=1}^{r} d\mu(\lambda_i), \tag{8}$$

where, writing $n = m - 2r$,

$$\begin{aligned} \mathcal{P}_{m,r}(q; \lambda_1, \ldots, \lambda_r) = \frac{m}{r} &\left(h_n(p^{\times r}, -\lambda_1, \ldots, -\lambda_r) + h_n(q^{\times r}, \lambda_1, \ldots, \lambda_r) \right) \\ &- h_{n-1}(q^{\times (r+1)}, \lambda_1, \ldots, \lambda_r). \end{aligned} \tag{9}$$

Proof. Put

$$Y_W(z) = \frac{z^2 R_W(z)}{1 - pz}, \qquad Y_U(z) = \frac{z^2 R_U(z)}{1 - qz}.$$

From $-\log(1 - Y) = \sum_{r \ge 1} Y^r / r$,

$$m[z^m]\mathcal{L}_W = \sum_{r \ge 1} \frac{m}{r} [z^{m-2r}] \frac{R_W(z)^r}{(1 - pz)^r}.$$

Since

$$R_W(z)^r = \int \cdots \int \prod_{i=1}^{r} (1 + \lambda_i z)^{-1} \prod_i d\mu(\lambda_i),$$

the rth term on the W side is the integral of $\frac{m}{r}h_n(p^{\times r}, -\lambda_1, \dots, -\lambda_r)$. The same argument on the U side gives the integral of $\frac{m}{r}h_n(q^{\times r}, \lambda_1, \dots, \lambda_r)$.

For the path term, using (7),

$$\frac{1}{1 - qz - z^2 R_U(z)} = \sum_{r \geq 0} \frac{(z^2 R_U(z))^r}{(1 - qz)^{r+1}}.$$

The $r = 0$ contribution to x_{m-1} is q^{m-1}, which cancels the leading term in (6). For $r \geq 1$, extracting $[z^{m-1}]$ gives the integral of

$$h_{n-1}(q^{\times (r+1)}, \lambda_1, \dots, \lambda_r).$$

Combining the three contributions proves the formula. $\square$

Corollary 4.2 (Master sufficient criterion)**.** *If $\mathcal{P}_{m,r}(q; \lambda_1, \dots, \lambda_r) \geq 0$ for every $1 \leq r \leq (m-1)/2$ and every $\lambda_i \in [-1/2, 1/2]$, then Conjecture 1.1 holds for this m and this value of q.*

5 Mixture of diagonals

The next theorem is the main structural simplification. It says that the multivariate kernel is an average of diagonal kernels.

Theorem 5.1 (Mixture-of-diagonals theorem)**.** *Let $(\Theta_1, \dots, \Theta_r) \sim \mathrm{Dir}(1, \dots, 1)$, the uniform distribution on the simplex. Define*

$$\widetilde{\mathcal{P}}_{m,r}(q, \ell) := \mathcal{P}_{m,r}(q; \ell, \dots, \ell).$$

Then

$$\mathcal{P}_{m,r}(q; \lambda_1, \dots, \lambda_r) = \mathbb{E}\left[\widetilde{\mathcal{P}}_{m,r}\left(q, \sum_{i=1}^{r} \Theta_i \lambda_i\right)\right]. \tag{10}$$

Consequently

$$\min_{\lambda_1, \dots, \lambda_r \in [-1/2, 1/2]} \mathcal{P}_{m,r}(q; \lambda_1, \dots, \lambda_r) = \min_{\ell \in [-1/2, 1/2]} \widetilde{\mathcal{P}}_{m,r}(q, \ell).$$

Proof. We use two elementary identities. First,

$$h_d(a^{\times k}, b_1, \dots, b_r) = \sum_{j=0}^{d} \binom{d - j + k - 1}{k - 1} a^{d-j} h_j(b_1, \dots, b_r).$$

Second, for $\Theta \sim \mathrm{Dir}(1^r)$,

$$\mathbb{E}(\Theta_1 \lambda_1 + \cdots + \Theta_r \lambda_r)^j = \frac{h_j(\lambda_1, \dots, \lambda_r)}{\binom{j+r-1}{r-1}}. \tag{11}$$

This is the standard Dirichlet moment formula, or equivalently the generating-function identity obtained by integrating powers over the simplex.

Apply the first identity to the three complete homogeneous polynomials in (9). For example,

$$h_n(p^{\times r}, -\vec{\lambda}) = \sum_{j=0}^{n} \binom{n - j + r - 1}{r - 1} p^{n-j} (-1)^j h_j(\vec{\lambda}).$$

On the diagonal with ℓ repeated r times,

$$h_n(p^{\times r},(-\ell)^{\times r})=\sum_{j=0}^{n}\binom{n-j+r-1}{r-1}p^{n-j}\binom{j+r-1}{r-1}(-\ell)^j.$$

Taking expectation with $\ell=\sum_i \Theta_i\lambda_i$ and using (11) recovers the previous expression. The same argument applies to $h_n(q^{\times r},\vec{\lambda})$ and to $h_{n-1}(q^{\times(r+1)},\vec{\lambda})$; in the last case the coefficient of ℓ^j on the diagonal is still multiplied by $\binom{j+r-1}{r-1}$, which is cancelled by (11). Linearity gives (10).

The minimum equality follows because every convex combination $\sum_i \Theta_i\lambda_i$ lies in the interval spanned by the λ_i, while diagonal choices $\lambda_1=\cdots=\lambda_r=\ell$ are allowed. □

Corollary 5.2 (Diagonal sufficient criterion)**.** *It is enough to prove*

$$\widetilde{\mathcal{P}}_{m,r}(q,\ell)\geq 0\qquad (1\leq r\leq (m-1)/2,\ -1/2\leq \ell\leq 1/2).$$

6 The diagonal family as a one-dimensional integral

Fix $r\geq 1$ and odd $n\geq 1$, and put $m=n+2r$. The diagonal family has a useful probabilistic form. Let $\Xi\sim\mathrm{Beta}(r,r)$ and set

$$V_x=qx+\ell(1-x),\qquad W_x=px-\ell(1-x).$$

Then $V_x+W_x=x$, and a direct use of the Dirichlet formula gives

$$\widetilde{\mathcal{P}}_{m,r}(q,\ell)=\binom{n+2r-1}{2r-1}\frac{n}{r}\left[\frac{m}{n}\mathbb{E}(V_\Xi^n+W_\Xi^n)-\mathbb{E}(\Xi V_\Xi^{n-1})\right].\tag{12}$$

Define

$$\rho_{n,m}(u)=\frac{m}{n}\big(u^n+(1-u)^n\big)-u^{n-1}.\tag{13}$$

Proposition 6.1 (One-dimensional kernel form)**.** *For $\ell>0$,*

$$\widetilde{\mathcal{P}}_{m,r}(q,\ell)=C_{m,r}\ell^{n+r}\int_0^\infty \frac{s^{r-1}}{(\ell+s)^m}\rho_{n,m}(q+s)\,ds,\tag{14}$$

where

$$C_{m,r}=\binom{n+2r-1}{2r-1}\frac{n}{r}\frac{\Gamma(2r)}{\Gamma(r)^2}>0.$$

Proof. In (12), write the $\mathrm{Beta}(r,r)$ density as

$$\frac{\Gamma(2r)}{\Gamma(r)^2}x^{r-1}(1-x)^{r-1}\,dx.$$

For $\ell>0$, set

$$u=\frac{V_x}{x}=q+\ell\frac{1-x}{x}=q+s,\qquad x=\frac{\ell}{\ell+s}.$$

Then $W_x=x(1-u)$, and

$$dx=-\frac{\ell}{(\ell+s)^2}\,ds,\qquad x^{n+r-1}(1-x)^{r-1}dx=-\ell^{n+r}s^{r-1}(\ell+s)^{-m}\,ds.$$

Changing variables from $x\in(0,1)$ to $s\in(\infty,0)$ gives (14). □

7 The sign structure of ρ

Lemma 7.1 (ρ-lemma)**.** *Let $n \geq 1$ be odd and $m = n + 2r \geq n + 2$. Then:*

(i)
$$\rho(u) = \frac{m}{n}(1-u)\big((1-u)^{n-1} - u^{n-1}\big) + \left(\frac{m}{n} - 1\right) u^{n-1},$$

and also
$$\rho(u) = \frac{m}{n}(1-u)^n + u^{n-1}\left(\frac{m}{n}u - 1\right).$$

(ii) $\rho(u) \geq 0$ *for* $u \notin (1/2, n/m)$.

(iii) *For every real u,*
$$\rho(u) + \rho(1-u) \geq 0.$$

More precisely,
$$\rho(u) + \rho(1-u) \geq (2u-1)\big(u^{n-1} - (1-u)^{n-1}\big) \geq 0.$$

(iv) *If $m \geq 2n$, then $\rho \geq 0$ on all of $\mathbb{R}$.*

(v) *On the window $u \in (1/2, n/m)$,*
$$-\rho(u) \leq u^{n-1}\left(1 - \frac{m}{n}u\right).$$

Proof. The two formulae in (i) are algebraic rearrangements of (13). For $u \leq 1/2$, the first formula is nonnegative because $n-1$ is even and $|1-u| \geq |u|$. For $n/m \leq u \leq 1$, the second formula is nonnegative. If $u \geq 1$, then $u^n - (u-1)^n \geq u^{n-1}$, so $\rho(u) \geq (m/n - 1)u^{n-1} \geq 0$. If $u = -v \leq 0$, then $(1+v)^n - v^n \geq nv^{n-1}$, so $\rho(u) \geq (m-1)v^{n-1} \geq 0$. This proves (ii); (iv) follows because the possible negative window is empty when $n/m \leq 1/2$.

For reflection,
$$\rho(u) + \rho(1-u) = \frac{2m}{n}(u^n + (1-u)^n) - u^{n-1} - (1-u)^{n-1}.$$

For odd n, $u^n + (1-u)^n \geq 0$ for every real u, and $m/n \geq 1$. Hence this is at least
$$2u^n + 2(1-u)^n - u^{n-1} - (1-u)^{n-1} = (2u-1)\big(u^{n-1} - (1-u)^{n-1}\big).$$

The final product is nonnegative because $n-1$ is even and $|u| \geq |1-u|$ exactly when $u \geq 1/2$. Finally, (v) follows by dropping the nonnegative term $(m/n)(1-u)^n$ from the second formula in (i). □

8 General positivity regimes

Theorem 8.1 (Pointwise regimes)**.** *For every $q \in [0, 1/2]$, $\widetilde{\mathcal{P}}_{m,r}(q, \ell) \geq 0$ in each of the following regimes:*

(a) $2r \geq n$;

(b) $\ell \leq 0$.

Proof. If $\ell > 0$ and $2r \geq n$, then $m = n + 2r \geq 2n$, so the kernel representation and Lemma 7.1(iv) give nonnegativity. If $\ell \leq 0$, use (12). For $x \in (0, 1]$,

$$\frac{V_x}{x} = q + \ell \frac{1-x}{x} \leq q \leq 1/2,$$

so the corresponding $x^n \rho(V_x/x)$ integrand is pointwise nonnegative by Lemma 7.1(ii). This proves the theorem. □

Theorem 8.2 (Integration-by-parts regime)**.** *For every $q \in [0, 1/2]$, if*

$$\ell \geq q + \frac{r}{m},$$

then $\widetilde{\mathcal{P}}_{m,r}(q, \ell) \geq 0$.

Proof. It is enough to prove the bracket in (12) is nonnegative. Assume $\ell > q$, since otherwise this theorem is contained in Theorem 8.1. The function $V(x) = \ell + (q - \ell)x$ is positive on $[0, 1]$.

First, $\mathbb{E}W_\Xi^n \geq 0$. Indeed, $W(x) = (p + \ell)(x - x_w)$ with $x_w = \ell/(p + \ell) \leq 1/2$, and the Beta$(r, r)$ law is symmetric about $1/2$. Pairing $x = 1/2 + t$ with $x = 1/2 - t$, the two values of $x - x_w$ have nonnegative sum and the larger absolute value is the positive one, so their odd nth powers sum to a nonnegative number.

Second, we prove

$$\mathbb{E}[\Xi V_\Xi^{n-1}] \leq \frac{r}{n(\ell - q)} \mathbb{E}[V_\Xi^n]. \tag{15}$$

For $r \geq 2$, integrate by parts against the density $c_r x^{r-1}(1-x)^{r-1}$; the boundary terms vanish and

$$\mathbb{E}[\Xi V^{n-1}] = \frac{c_r}{n(\ell - q)} \int_0^1 x^{r-1}(1-x)^{r-2}\big(r - (2r-1)x\big)V(x)^n \, dx.$$

Discarding the negative part of the integrand and using

$$\frac{r - (2r-1)x}{1-x} \leq r \qquad (0 \leq x \leq r/(2r-1))$$

gives (15). For $r = 1$, integration by parts gives directly

$$\mathbb{E}[\Xi V^{n-1}] = \frac{\mathbb{E}[V^n] - q^n}{n(\ell - q)} \leq \frac{\mathbb{E}[V^n]}{n(\ell - q)},$$

which is (15) for $r = 1$.

If $\ell \geq q + r/m$, then $m/n \geq r/(n(\ell - q))$. Thus

$$\frac{m}{n}\mathbb{E}[V^n + W^n] \geq \frac{m}{n}\mathbb{E}[V^n] \geq \mathbb{E}[\Xi V^{n-1}],$$

which proves the claim. □

9 The repaired all-length theorem for $r = 1$

The following proof is the main correction to Claude's note. The issue in the original proof was that the reflection step paired negative mass in the interval $(s_a, 2s_a)$ with positive mass in $(0, s_a)$ and then the strip estimate attempted to reuse part of the same positive mass. The repair is to pair only the middle band $(2\varepsilon, 3\varepsilon)$ with $(\varepsilon, 2\varepsilon)$ and reserve $(0, \varepsilon)$ for the surplus estimate.

Theorem 9.1 ($r = 1$ diagonal positivity, all lengths)**.** *Let $r = 1$, $m = n + 2$ with $n \geq 3$ odd, $q \in [0, 1/3]$, and $\ell \in [-1/2, 1/2]$. Then*

$$\widetilde{\mathcal{P}}_{m,1}(q, \ell) \geq 0.$$

Proof. The cases $\ell \leq 0$ and $\ell \geq q + 1/m$ are covered by Theorems 8.1 and 8.2. We assume from now on that

$$0 < \ell < q + \frac{1}{m}.$$

By Proposition 6.1, it suffices to prove

$$I := \int_0^\infty \frac{\rho(q+s)}{(\ell+s)^m}\, ds \geq 0, \qquad \rho = \rho_{n,m}.$$

The case $m = 5$ is elementary: directly from (9),

$$\widetilde{\mathcal{P}}_{5,1}(q, \ell) = 4\ell^2 + (8q - 5)\ell + 12q^2 - 15q + 5.$$

On $0 \leq q \leq 1/3$ and $-1/2 \leq \ell \leq 1/2$, this quadratic is nonnegative. Indeed, if its vertex $\ell_* = 5/8 - q$ lies in the interval, the minimum is

$$\frac{128q^2 - 160q + 55}{16} \geq \frac{143}{144} > 0;$$

otherwise the minimum over the interval is attained at an endpoint, where direct substitution gives positive quadratics. We may therefore assume $m \geq 7$, hence $n \geq 5$.

Put

$$\varepsilon = \frac{1-2q}{4}, \qquad s_a = \frac{1}{2} - q = 2\varepsilon, \qquad \sigma = 3\varepsilon, \qquad s_b = \frac{n}{m} - q.$$

By Lemma 7.1, the integrand can be negative only for $s \in (s_a, s_b)$.

Disjoint reflection band. For $s \in (s_a, \sigma) = (2\varepsilon, 3\varepsilon)$ define

$$\widehat{s} = 2s_a - s = 4\varepsilon - s \in (\varepsilon, 2\varepsilon).$$

Then $q + \widehat{s} = 1 - (q + s)$. Since $(\ell + s)^{-m}$ is decreasing in s,

$$(\ell + \widehat{s})^{-m} \geq (\ell + s)^{-m}.$$

Together with Lemma 7.1(iii), this gives

$$\frac{\rho(q+s)}{(\ell+s)^m} + \frac{\rho(q+\widehat{s})}{(\ell+\widehat{s})^m} \geq 0.$$

Thus the negative contribution on $(2\varepsilon, 3\varepsilon)$ is paid for by the disjoint positive partner band $(\varepsilon, 2\varepsilon)$.

It remains only to compare the possible deficit on $(3\varepsilon, s_b)$ with the unused surplus on $(0, \varepsilon)$. If $s_b \leq 3\varepsilon$, there is no remaining deficit and we are done. Assume $s_b > 3\varepsilon$, and define

$$\eta = s_b - 3\varepsilon = \frac{1+2q}{4} - \frac{2}{m} > 0.$$

For $s \in (3\varepsilon, s_b)$ set $V = q + s$. Then $V \in (p - \varepsilon, n/m)$, and Lemma 7.1(v) gives

$$-\rho(V) \leq V^{n-1}\left(1 - \frac{m}{n}V\right).$$

The function $V^{n-1}(\ell+V-q)^{-m}$ is decreasing on this interval because

$$\frac{d}{dV}\log\frac{V^{n-1}}{(\ell+V-q)^m}<0 \iff (n-1)(\ell-q)<3V,$$

and the left side is $<(n-1)/m<1$, while $V\geq p-\varepsilon\geq 7/12$. The bracket $1-(m/n)V$ is also decreasing. Hence the total remaining deficit D is bounded by its left-endpoint upper sum:

$$D\leq\frac{m}{n}\,\eta^2\,\frac{(p-\varepsilon)^{n-1}}{(\ell+3\varepsilon)^m}. \tag{16}$$

Indeed, $1-(m/n)(p-\varepsilon)=m\eta/n$ and the interval length is η.

On the surplus band $(0,\varepsilon)$, we have $q+s\leq q+\varepsilon\leq 5/12<1/2$. Using the second grouping of Lemma 7.1(i),

$$\rho(q+s)\geq\frac{m}{n}(p-s)^n-(q+s)^{n-1}\geq\frac{m}{n}(p-\varepsilon)^n c_n,$$

where

$$c_n=1-\frac{n}{m}\frac{(q+\varepsilon)^{n-1}}{(p-\varepsilon)^n}.$$

Since $(q+\varepsilon)/(p-\varepsilon)\leq 5/7$ and $(p-\varepsilon)^{-1}\leq 12/7$, for $n\geq 5$,

$$c_n\geq 1-\frac{12}{7}\left(\frac{5}{7}\right)^{n-1}\geq 1-\frac{12}{7}\left(\frac{5}{7}\right)^4>\frac{1}{2}.$$

Therefore the unused surplus Σ on $(0,\varepsilon)$ satisfies

$$\Sigma\geq\frac{m}{n}\,(p-\varepsilon)^n c_n\frac{\varepsilon}{(\ell+\varepsilon)^m}. \tag{17}$$

It remains to prove $\Sigma\geq D$. From (16) and (17), it is enough that

$$\varepsilon c_n(p-\varepsilon)\left(\frac{\ell+3\varepsilon}{\ell+\varepsilon}\right)^m\geq\eta^2. \tag{18}$$

Now

$$\varepsilon\geq\frac{1}{12},\qquad p-\varepsilon\geq\frac{7}{12},\qquad c_n>\frac{1}{2}.$$

Also $\ell<q+1/m$, and the function $t\mapsto(a+t)/(b+t)$ is decreasing when $a>b$, so

$$\frac{\ell+3\varepsilon}{\ell+\varepsilon}\geq\frac{q+1/m+3\varepsilon}{q+1/m+\varepsilon}.$$

The right-hand side is decreasing in both q and $1/m$; hence for $q\leq 1/3$ and $m\geq 7$ it is at least

$$\frac{1/3+1/7+3/12}{1/3+1/7+1/12}=\frac{61}{47}.$$

Thus the left side of (18) is at least

$$\frac{7}{288}\left(\frac{61}{47}\right)^m.$$

For $m=7$, we have $\eta\leq 5/12-2/7=11/84$, and

$$\frac{7}{288}\left(\frac{61}{47}\right)^7>\left(\frac{11}{84}\right)^2.$$

For $m \ge 9$, we use $\eta \le 5/12$ and

$$\frac{7}{288}\left(\frac{61}{47}\right)^9 > \left(\frac{5}{12}\right)^2.$$

This proves (18). Summing the disjoint reflection pair, the unused surplus, the remaining deficit, and the regions where $\rho \ge 0$, we get $I \ge 0$. $\square$

10 The residual strip and exact finite verification

After Theorems 8.1, 8.2, and 9.1, the only diagonal cases left for $q \le 1/3$ are

$$r \ge 2, \qquad n > 2r, \qquad 0 < \ell < q + \frac{r}{m}. \tag{19}$$

For these cases, the one-dimensional integral (14) has a negative window, but the following criterion gives a rigorous way to pay for it by reflection plus two positive surplus bands.

Theorem 10.1 (Two-sided surplus criterion)**.** *Let $r \ge 2$, $n > 2r$ be odd, $m = n + 2r$, $q \in [0, 1/2)$, and $\ell > 0$. Put*

$$\kappa(s) = s^{r-1}(\ell + s)^{-m}, \qquad s_a = \frac{1}{2} - q, \qquad s_b = \frac{n}{m} - q, \qquad \nu = \frac{n}{m}.$$

Assume $s_a < s_b$ and

$$(n-1)(\ell - q) \le \frac{2r+1}{2}. \tag{20}$$

Let $\sigma \in [s_a, \min(2s_a, s_b)]$ satisfy

$$\kappa(2s_a - s) \ge \kappa(s) \qquad (s_a \le s \le \sigma). \tag{21}$$

Choose partitions

$$\sigma = t_0 < t_1 < \cdots < t_K = s_b,$$

$$0 = a_0 < a_1 < \cdots < a_J \le \min(2s_a - \sigma, s_a),$$

$$0 = d_0 < d_1 < \cdots < d_L \le \frac{2r}{m}.$$

Define

$$\overline{\Delta} := s_b^{r-1} \sum_{i=0}^{K-1} (t_{i+1} - t_i)(q + t_i)^{n-1} \left(1 - \frac{m}{n}(q + t_i)\right)_+ (\ell + t_i)^{-m},$$

$$\Sigma_L := \frac{m}{n} \sum_{j=1}^{J-1} (a_{j+1} - a_j) a_j^{r-1} (p - a_{j+1})^n \left(1 - \frac{n}{m} \frac{(q + a_{j+1})^{n-1}}{(p - a_{j+1})^n}\right)_+ (\ell + a_{j+1})^{-m},$$

$$\Sigma_R := \frac{m}{n} \sum_{i=1}^{L-1} d_i (d_{i+1} - d_i)(\nu + d_i)^{n-1} (s_b + d_i)^{r-1} (\ell + s_b + d_{i+1})^{-m}.$$

If

$$\Sigma_L + \Sigma_R \ge \overline{\Delta},$$

then

$$\int_0^\infty \kappa(s) \rho_{n,m}(q + s)\, ds \ge 0.$$

Proof. By Lemma 7.1, the integrand is nonnegative outside (s_a, s_b). For $s \in (s_a, \sigma)$, pair it with $\widehat{s} = 2s_a - s$. Then $q + \widehat{s} = 1 - (q + s)$, and (21) plus Lemma 7.1(iii) gives

$$\kappa(s)\rho(q+s) + \kappa(\widehat{s})\rho(q+\widehat{s}) \geq 0.$$

Thus the negative part on (s_a, σ) is paid for by the disjoint partner interval $(2s_a - \sigma, s_a)$.

It remains to bound the deficit on $D = (\sigma, s_b)$. By Lemma 7.1(v),

$$-\rho(q+s) \leq (q+s)^{n-1}\left(1 - \frac{m}{n}(q+s)\right)_+.$$

The factor $(q+s)^{n-1}(\ell+s)^{-m}$ is decreasing on D: writing $V = q+s > 1/2$, its logarithmic derivative is negative whenever

$$(n-1)(\ell - q) < (m-n+1)V = (2r+1)V,$$

which follows from (20). The bracket is decreasing as well, and $s^{r-1} \leq s_b^{r-1}$. Hence the deficit on each interval $[t_i, t_{i+1}]$ is bounded by the corresponding term of $\overline{\Delta}$.

For the left surplus band, $a_J \leq 2s_a - \sigma$ makes it disjoint from the partner interval used above. On each $[a_j, a_{j+1}]$, the second formula in Lemma 7.1(i) and monotonicity of the factors give the lower sum Σ_L. For the right surplus band, $q+s \in [n/m, 1]$ on $[s_b, s_b + 2r/m]$, and the same formula gives the lower sum Σ_R. All unused regions are nonnegative, so $\Sigma_L + \Sigma_R \geq \overline{\Delta}$ implies the integral is nonnegative. □

Definition 10.2 (Exact strip certificate)**.** *For an odd integer $M \geq 5$, say that* StripCert(M) *holds if, for every odd $m \leq M$ and every residual pair (m, r) satisfying (19), the exact rational interval checker supplied with this note proves the hypotheses of Theorem 10.1 for all $(q, \ell) \in [0, 1/3] \times [0, 1/2]$.*

Proposition 10.3 (Verified finite certificate)**.** *The accompanying script* `clean_two_sided_shift_checker.py` *verifies* StripCert(61) *using exact rational interval arithmetic. The run log included as* `clean_checker_m61.log` *records successful verification for all residual pairs with odd $5 \leq m \leq 61$.*

Proof status. This is a finite, deterministic computer-assisted certificate. The script uses rational arithmetic from Python's `fractions.Fraction`. Each interval cell is certified by one of the analytic regimes above or by rational upper and lower Riemann sums appearing in Theorem 10.1. No floating-point inequality is used for Proposition 10.3. The code is short enough to audit directly; the log records the exact list of m values checked. Claude's original checker also advertised a recorded run to $M = 201$, but no certificate log was supplied in the uploaded material, so this note does not use 201 as a proved bound. □

11 Analytic closure of the residual strip for $q \leq 1/3$

We now close the residual strip for all odd lengths. This is the new ingredient beyond the finite certificate in Proposition 10.3. Throughout this section assume

$$q \leq \frac{1}{3}, \qquad m = n + 2r \geq 63, \qquad r \geq 2, \qquad n > 2r, \qquad 0 < \ell < q + \frac{r}{m}. \tag{22}$$

Put

$$\theta = \frac{r}{m}, \qquad \nu = \frac{n}{m} = 1 - 2\theta, \qquad a = \frac{1}{2} - q, \qquad b = \nu - q, \qquad L = b - a = \nu - \frac{1}{2} = \frac{1}{2} - 2\theta,$$

and

$$\varepsilon = \frac{1-2q}{4} = \frac{a}{2}, \qquad p = 1-q, \qquad \kappa(s) = \frac{s^{r-1}}{(\ell+s)^m}.$$

The negative window of the kernel integral in Proposition 6.1 is exactly

$$q+s \in \left(\frac{1}{2}, \nu\right), \qquad \text{that is,} \qquad s \in (a,b).$$

Lemma 11.1 (One-sided bounds for the kernel sign function)**.** *For $\rho = \rho_{n,m}$ and $\nu = n/m$,*

$$-\rho(u) \le \frac{\nu-u}{\nu} u^{n-1} \qquad \textit{for } 1/2 < u < \nu, \tag{23}$$

$$\rho(u) \ge \frac{u-\nu}{\nu} u^{n-1} \qquad \textit{for } u > \nu, \tag{24}$$

$$\rho(u) \ge \frac{m}{n}(1-u)^n - u^{n-1} \qquad \textit{for all } u. \tag{25}$$

Proof. Since

$$\nu\rho(u) = u^n + (1-u)^n - \nu u^{n-1},$$

we have, for $u < \nu$,

$$-\nu\rho(u) = (\nu-u)u^{n-1} - (1-u)^n \le (\nu-u)u^{n-1},$$

and, for $u > \nu$,

$$\nu\rho(u) = (u-\nu)u^{n-1} + (1-u)^n \ge (u-\nu)u^{n-1}.$$

The final inequality is obtained from the definition of ρ by discarding the nonnegative term $(m/n)u^n$. ☐

Lemma 11.2 (Left-surplus estimate)**.** *Under the assumptions* (22)*, the integral*

$$I = \int_0^\infty \kappa(s)\rho_{n,m}(q+s)\,ds$$

is nonnegative in each of the following two cases:

(a) $0 < \theta \le 1/6$*;*

(b) $1/6 \le \theta < 1/4$ *and* $0 < \ell \le 2/5$.

Proof. The only possible negative contribution is from $s \in (a,b)$. On this interval, Lemma 11.1 gives

$$-\rho(q+s) \le (q+s)^{n-1}\left(1 - \frac{q+s}{\nu}\right).$$

The function

$$s \mapsto \frac{(q+s)^{n-1}}{(\ell+s)^m}$$

is decreasing on (a,b). Indeed, its logarithmic derivative is nonpositive once

$$(n-1)(\ell-q) \le (2r+1)(q+s),$$

and this follows from $\ell < q + r/m$ and $q+s \ge 1/2$. Therefore the total possible deficit D obeys

$$D \le b^{r-1}\frac{(1/2)^{n-1}}{(\ell+a)^m}\frac{L^2}{2\nu}. \tag{26}$$

On $0 \le s \le \varepsilon$, Lemma 11.1 gives

$$\rho(q+s) \ge \frac{m}{n}(p-s)^n - (q+s)^{n-1} \ge \frac{m}{n}(p-\varepsilon)^n c_n,$$

where

$$c_n = 1 - \nu \frac{(q+\varepsilon)^{n-1}}{(p-\varepsilon)^n}.$$

Since $q+\varepsilon \le 5/12$, $p-\varepsilon \ge 7/12$, and in (22) one has $n \ge 33$, we get

$$c_n \ge 1 - \frac{12}{7}\left(\frac{5}{7}\right)^{32} > \frac{99}{100}.$$

The positive contribution Σ from $(0,\varepsilon)$ therefore satisfies

$$\Sigma \ge \frac{m}{n}(p-\varepsilon)^n c_n \frac{\varepsilon^r}{r(\ell+\varepsilon)^m}. \tag{27}$$

Dividing (27) by (26) gives

$$\frac{\Sigma}{D} \ge c_n \frac{b}{rL^2}\left(2(p-\varepsilon)\right)^n \left(\frac{\varepsilon}{b}\right)^r \left(\frac{\ell+a}{\ell+\varepsilon}\right)^m. \tag{28}$$

Case (a): $0 < \theta \le 1/6$. Here $\ell < q+\theta$. The elementary monotonicities in q and ℓ give

$$2(p-\varepsilon) \ge \frac{7}{6}, \qquad \frac{\varepsilon}{b} \ge \frac{1}{8-24\theta},$$
$$\frac{\ell+a}{\ell+\varepsilon} \ge \frac{1/2+\theta}{5/12+\theta}, \qquad \frac{b}{L^2} \ge \frac{2/3-2\theta}{(1/2-2\theta)^2}.$$

Thus

$$\frac{\Sigma}{D} \ge \frac{99}{100m}\frac{2/3-2\theta}{\theta(1/2-2\theta)^2}\left[\left(\frac{7}{6}\right)^{1-2\theta}(8-24\theta)^{-\theta}\frac{1/2+\theta}{5/12+\theta}\right]^m. \tag{29}$$

Appendix A proves that the right-hand side is at least 1 for every integer pair in this case. Hence $\Sigma \ge D$.

Case (b): $1/6 \le \theta < 1/4$ *and* $\ell \le 2/5$. The same comparison yields

$$2(p-\varepsilon) \ge \frac{7}{6}, \qquad \frac{\varepsilon}{b} \ge \frac{1}{8-24\theta}, \qquad \frac{\ell+a}{\ell+\varepsilon} \ge \frac{34}{29},$$

where the last inequality follows by taking $q = 1/3$ and $\ell = 2/5$. Therefore

$$\frac{\Sigma}{D} \ge \frac{99}{100m}\frac{2/3-2\theta}{\theta(1/2-2\theta)^2}\left[\left(\frac{7}{6}\right)^{1-2\theta}(8-24\theta)^{-\theta}\frac{34}{29}\right]^m. \tag{30}$$

Appendix A proves that this lower bound is also at least 1. Hence $\Sigma \ge D$.

Outside (a,b) the integrand is nonnegative. Thus in both cases $I \ge 0$. $\square$

Lemma 11.3 (The upper-reflection threshold)**.** *Let* $m \ge 63$ *be odd,* $r/m \ge 1/6$*,* $n = m-2r$*,* $\nu = n/m$*, and* $n > 2r$*. For every* $0 < b \le \nu$ *define*

$$L_A(b) = \frac{m}{(r-1)/b + (n-1)/\nu} - b.$$

Then $L_A(b) \le 2/5$.

Proof. Put $A = r - 1$ and $B = (n-1)/\nu$. Since

$$L_A(b) = \frac{mb}{A + Bb} - b,$$

maximising over $b > 0$ gives

$$L_A(b) \le \frac{(\sqrt{m} - \sqrt{A})^2}{B}.$$

Because $m \ge 63$ is odd and $r/m \ge 1/6$, a check of residues modulo 6 gives

$$\frac{r-1}{m} \ge \frac{2}{13}.$$

Also $n > 2r$ gives $\nu > 1/2$, so

$$B = \frac{n-1}{\nu} = m - \frac{1}{\nu} > m - 2.$$

Therefore

$$L_A(b) \le \frac{m}{m-2}\left(1 - \sqrt{\frac{2}{13}}\right)^2 \le \frac{63}{61}\left(1 - \sqrt{\frac{2}{13}}\right)^2 < \frac{2}{5}.$$

The final inequality is equivalent, after moving positive terms and squaring once, to a positive rational inequality. $\square$

Lemma 11.4 (Upper-end reflection)**.** *Under the assumptions* (22)*, if $\theta \ge 1/6$ and $\ell > 2/5$, then*

$$\int_0^\infty \kappa(s)\rho_{n,m}(q+s)\,ds \ge 0.$$

Proof. Write the negative point as $u = \nu - e$, where $0 < e < L = \nu - 1/2$, and reflect it through the upper endpoint ν to $u' = \nu + e$. In s-coordinates this pairs $s = b - e$ with $s' = b + e$. Since $\theta \ge 1/6$, we have $\nu \le 2/3$, and hence $u' \le 2\nu - 1/2 \le 1$.

By Lemma 11.1,

$$-\rho(\nu - e) \le \frac{e}{\nu}(\nu - e)^{n-1}, \qquad \rho(\nu + e) \ge \frac{e}{\nu}(\nu + e)^{n-1}.$$

Thus pointwise payment follows if

$$\left(\frac{b+e}{b-e}\right)^{r-1}\left(\frac{\nu+e}{\nu-e}\right)^{n-1} \ge \left(\frac{\ell+b+e}{\ell+b-e}\right)^m. \tag{31}$$

Put $C = \ell + b$. By Lemma 11.3 and $\ell > 2/5$,

$$\frac{r-1}{b} + \frac{n-1}{\nu} \ge \frac{m}{C}.$$

Taking logarithms in (31), expanding

$$\log\frac{X+e}{X-e} = 2\sum_{j\ge 0}\frac{e^{2j+1}}{(2j+1)X^{2j+1}}, \qquad 0 < e < X,$$

and using $b \le \nu < C$, each coefficient is bounded below by

$$C^{-2j}\left(\frac{r-1}{b} + \frac{n-1}{\nu} - \frac{m}{C}\right) \ge 0.$$

Hence (31) holds. The reflected positive mass pays the whole negative window pointwise; all remaining parts of the integral are nonnegative by Lemma 7.1. This proves the claim. $\square$

Proposition 11.5 (Residual diagonal strip, all odd lengths)**.** *Let $q \le 1/3$, let $m = n + 2r$ be odd, and assume*

$$r \ge 2, \qquad n > 2r, \qquad 0 < \ell < q + \frac{r}{m}.$$

Then

$$\widetilde{\mathcal{P}}_{m,r}(q, \ell) \ge 0.$$

Proof. For odd $m \le 61$, this is Proposition 10.3. Let $m \ge 63$ and put $\theta = r/m$. If $\theta \le 1/6$, Lemma 11.2(a) applies. If $\theta \ge 1/6$ and $\ell \le 2/5$, Lemma 11.2(b) applies. If $\theta \ge 1/6$ and $\ell > 2/5$, Lemma 11.4 applies. These alternatives cover the residual strip. □

12 The high-density theorem

Theorem 12.1 (High-density odd-cycle lower bound)**.** *Let W be a graphon with edge density $p \ge 2/3$. Then for every odd $m \ge 3$,*

$$t(C_m, W) \ge p^m - p(1-p)^{m-1}.$$

Proof. Put $q = 1 - p \le 1/3$. By Corollary 5.2, it suffices to prove $\widetilde{\mathcal{P}}_{m,r}(q, \ell) \ge 0$ for every r and every $\ell \in [-1/2, 1/2]$. Theorem 8.1 covers $\ell \le 0$ and $2r \ge n$. Theorem 8.2 covers $\ell \ge q + r/m$. Theorem 9.1 covers $r = 1$. Proposition 11.5 covers the only remaining case. Therefore all diagonal kernels are nonnegative, hence all multivariate kernels are nonnegative by Theorem 5.1, hence $\Phi_m \ge 0$ by Theorem 4.1. The deletion inequality and the two-sided identity then imply the desired odd-cycle bound. □

13 A second theorem: rank-two complement graphons and all bipodal graphons

The high-density theorem applies to all graphons but only for $p \ge 2/3$. The spectral-shift method also gives a separate all-density theorem for every graphon whose complement has one mean-zero direction. This includes every two-block graphon, balanced or unbalanced.

Theorem 13.1 (Rank-two complement graphons are safe)**.** *Let W be a graphon and put $U = 1 - W$. Suppose that there are $q \in [0, 1]$, $u, \lambda \in \mathbb{R}$, and a measurable function ψ satisfying*

$$\int \psi = 0, \qquad \int \psi^2 = 1,$$

such that

$$U(x, y) = q + u(\psi(x) + \psi(y)) + \lambda\psi(x)\psi(y) \tag{32}$$

for almost every (x, y), with $0 \le U \le 1$. Let $p = 1 - q$. Then for every odd $m \ge 3$,

$$t(C_m, W) \ge p^m - p(1-p)^{m-1}.$$

Corollary 13.2 (Arbitrary bipodal graphons)**.** *The odd-cycle lower bound holds for every two-block graphon, with arbitrary block sizes and arbitrary block values in $[0, 1]$.*

Proof. A two-block graphon becomes two-dimensional after passing to block-constant functions. In the orthonormal basis consisting of $\mathbf{1}$ and the normalized mean-zero block vector, its complement has the form (32). Theorem 13.1 applies. □

We now prove Theorem 13.1. The case $p \le 1/2$ is immediate because the right-hand side is nonpositive, so assume $p > 1/2$. On $\mathrm{span}\{\mathbf{1}, \psi\}$ the nonzero part of T_U is

$$\begin{pmatrix} q & u \\ u & \lambda \end{pmatrix}.$$

Conjugating $T_W = J - T_U$ by the sign flip on ψ gives the isospectral matrix

$$M = \begin{pmatrix} p & u \\ u & -\lambda \end{pmatrix}, \qquad t(C_m, W) = \operatorname{Tr} M^m. \tag{33}$$

Lemma 13.3 (Mean-zero compression bound)**.** *If $0 \le U \le 1$ and $f \perp \mathbf{1}$, $\|f\|_2 = 1$, then*

$$-\frac{1}{2} \le \langle f, T_U f \rangle \le \frac{1}{2}.$$

In particular $|\lambda| \le 1/2$ in (32).

Proof. Write $f = f_+ - f_-$ and $a = \int f_+ = \int f_-$. Since $0 \le U \le 1$,

$$\langle f, T_U f \rangle \le \left(\int f_+ \right)^2 + \left(\int f_- \right)^2 = 2a^2.$$

If f_+ and f_- are supported on sets of measures α and β, Cauchy's inequality gives $a^2 \le \alpha \int f_+^2$ and $a^2 \le \beta \int f_-^2$. Hence

$$1 = \int f_+^2 + \int f_-^2 \ge a^2 \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) \ge 4a^2,$$

because $\alpha + \beta \le 1$. Thus $\langle f, T_U f \rangle \le 1/2$. Applying the same argument to $1 - U$ and using $\langle f, T_{1-U} f \rangle = -\langle f, T_U f \rangle$ gives the lower bound. $\square$

Lemma 13.4 (Linear spectral-shift bound)**.** *Assume $p \ge 1/2$, $|\lambda| \le p$, and $m \ge 3$ is odd. Then*

$$\operatorname{Tr} \begin{pmatrix} p & u \\ u & -\lambda \end{pmatrix}^m \ge p^m - \lambda^m + m u^2 \frac{p^{m-1} - \lambda^{m-1}}{p + \lambda},$$

with the quotient interpreted by continuity if $p + \lambda = 0$.

Proof. Schur's determinant formula gives

$$\det(I - zM) = (1 - pz)(1 + \lambda z) - u^2 z^2.$$

Therefore

$$-\log \det(I - zM) = -\log(1 - pz) - \log(1 + \lambda z) - \log \left(1 - \frac{u^2 z^2}{(1 - pz)(1 + \lambda z)} \right).$$

Taking $m[z^m]$ gives $\operatorname{Tr}(M^m)$. The coefficient of z^j in $((1 - pz)(1 + \lambda z))^{-1}$ is nonnegative when $|\lambda| \le p$. Keeping only the term linear in u^2 in the last logarithm gives the stated lower bound; since $m - 2$ is odd, the needed coefficient is $(p^{m-1} - \lambda^{m-1})/(p + \lambda)$. $\square$

If $\lambda \le q$, Theorem 13.1 follows immediately: if $\lambda < 0$, then $p^m - \lambda^m \ge p^m$, and if $0 \le \lambda \le q$, then $p^m - \lambda^m \ge p^m - q^m \ge p^m - pq^{m-1}$. Thus only the frontier case

$$p > 1/2, \qquad \lambda > q$$

remains.

Lemma 13.5 (Pointwise boundedness forces coupling). *In the frontier case,*

$$u^2 \geq \frac{p\lambda(\lambda - q)^2}{(p-\lambda)^2}.$$

Proof. Because $\lambda > 0$, the upper bound $U \leq 1$ implies that ψ is essentially bounded. Let

$$A = \operatorname{ess\,sup} \psi, \qquad B = -\operatorname{ess\,inf} \psi.$$

Mean zero and variance one imply $AB \geq 1$: integrate $(A - \psi)(\psi + B) \geq 0$. The inequalities below are first written at extremal values; the rigorous version follows by using essential ε-extrema and letting $\varepsilon \downarrow 0$.

From $U \geq 0$ at opposite extremes,

$$q + u(A - B) - \lambda AB \geq 0.$$

Since $\lambda > q$ and $AB \geq 1$, orient ψ so that $A > B$ and $u > 0$. From $U \leq 1$ at the positive extreme,

$$q + 2uA + \lambda A^2 \leq 1,$$

so $A \leq \sqrt{p/\lambda}$. Since $AB \geq 1$,

$$A - B \leq A - \frac{1}{A} \leq \sqrt{\frac{p}{\lambda}} - \sqrt{\frac{\lambda}{p}} = \frac{p - \lambda}{\sqrt{p\lambda}}.$$

Combining this with $u(A - B) \geq \lambda AB - q \geq \lambda - q$ gives the desired lower bound. □

Lemma 13.6 (Scalar domination). *For every odd $m \geq 3$ and all $0 \leq s \leq x < 1$,*

$$s^{m-1} - x^m + mx\frac{(x-s)^2}{(1-x)^2}\frac{1-x^{m-1}}{1+x} \geq 0.$$

Proof. Put $n = m - 1$, so n is even. The claim is equivalent to

$$s^n + A_n(x)(x-s)^2 \geq x^{n+1}, \qquad A_n(x) = \frac{(n+1)x(1-x^n)}{(1-x)^2(1+x)}.$$

Assume $0 < x < 1$ and write $s = x(1-z)$, $0 \leq z \leq 1$. After division by x^n, it is enough to show

$$(1-z)^n + B_n(x)z^2 \geq x, \qquad B_n(x) = \frac{(n+1)x^{3-n}(1-x^n)}{(1-x)^2(1+x)}.$$

For $n = 2$, the left side is minimized at value $3x/(1+2x) \geq x$. For $n \geq 4$, Bernoulli gives $(1-z)^n \geq 1 - nz$, so it suffices that $B_n(x) \geq n^2/(4(1-x))$. This is equivalent to

$$4(n+1)x^{3-n}\frac{1-x^n}{1-x^2} \geq n^2.$$

Writing $n = 2k$,

$$x^{3-n}\frac{1-x^n}{1-x^2} = x^{3-n} + x^{5-n} + \cdots + x \geq k - 1 = \frac{n}{2} - 1,$$

because the first $k-1$ exponents are nonpositive. Hence the left side is at least $2(n+1)(n-2) \geq n^2$ for even $n \geq 4$. □

Proof of Theorem 13.1. In the frontier case, Lemmas 13.4 and 13.5 give

$$\operatorname{Tr}(M^m) - \left(p^m - pq^{m-1}\right) \geq pq^{m-1} - \lambda^m \\ + m\frac{p\lambda(\lambda-q)^2}{(p-\lambda)^2}\frac{p^{m-1}-\lambda^{m-1}}{p+\lambda}.$$

Set $s = q/p$ and $x = \lambda/p$. Since $0 \leq s \leq x < 1$, division by p^m turns the right-hand side into the expression in Lemma 13.6. Thus $\operatorname{Tr}(M^m) \geq p^m - pq^{m-1}$, which is the desired inequality by (33). $\square$

14 Region II: what the new method says and does not say

Theorem 12.1 proves the conjecture for $q \leq 1/3$. For $1/3 < q < 1/2$, the trace-square budget implies that the complement compression A can have at most one eigenvalue above q. This is the one-frontier regime. Claude's note contains a useful aggregate forced-coupling lemma, which is rigorous and worth keeping.

Lemma 14.1 (Aggregate forced coupling)**.** *Let $\phi \in \mathbf{1}^\perp$ be a unit eigenfunction of A with eigenvalue $\alpha \geq 0$. Put*

$$a = \int \phi^+ = \int \phi^-,$$

and define

$$w^+ = (\phi^+ - a) - \langle \phi^+, \phi\rangle\phi, \qquad w^- = (\phi^- - a) - \langle \phi^-, \phi\rangle\phi.$$

Then

$$a\langle g, |\phi|\rangle \geq \alpha\|\phi^+\|_2^2\|\phi^-\|_2^2 - qa^2 - \langle w^+, Aw^-\rangle \geq \alpha\|\phi^+\|_2^2\|\phi^-\|_2^2 - qa^2 - \frac{1}{2}\|w^+\|_2\|w^-\|_2. \tag{34}$$

Consequently

$$a\|g\|_2\sqrt{1-4a^2} \geq \alpha\|\phi^+\|_2^2\|\phi^-\|_2^2 - qa^2 - \frac{1}{2}\|w^+\|_2\|w^-\|_2. \tag{35}$$

Proof. Write the doubly-centred kernel of A as $\mathcal{A}$, so that

$$U(x,y) = q + g(x) + g(y) + \mathcal{A}(x,y).$$

Testing the pointwise inequality $U \geq 0$ against the nonnegative function $\phi^+(x)\phi^-(y)$ gives

$$0 \leq qa^2 + a\langle g, \phi^+\rangle + a\langle g, \phi^-\rangle + \langle \phi^+ - a, A(\phi^- - a)\rangle.$$

Since $\langle g, \phi^+\rangle + \langle g, \phi^-\rangle = \langle g, |\phi|\rangle$, and

$$\langle \phi^+ - a, \phi\rangle = \|\phi^+\|_2^2, \qquad \langle \phi^- - a, \phi\rangle = -\|\phi^-\|_2^2,$$

the decomposition of $\phi^\pm - a$ into its ϕ component plus $w^\pm$ gives

$$\langle \phi^+ - a, A(\phi^- - a)\rangle = -\alpha\|\phi^+\|_2^2\|\phi^-\|_2^2 + \langle w^+, Aw^-\rangle.$$

Rearranging proves the first inequality in (34). The second follows from $\|A\| \leq 1/2$. Finally,

$$\langle g, |\phi|\rangle = \langle g, |\phi| - 2a\rangle \leq \|g\|_2\||\phi| - 2a\|_2 = \|g\|_2\sqrt{1-4a^2},$$

because $\int g = 0$, $\int |\phi| = 2a$, and $\int \phi^2 = 1$. $\square$

The aggregate lemma also explains why a tempting extension of the rank-one forced-coupling theorem is false.

Proposition 14.2 (Failure of the naive total-coupling bound)**.** *Let $s \in (0, 1/3)$, and let U_s be the graphon that is* 1 *on two disjoint diagonal cliques of measure* $(1-s)/2$ *each and* 0 *elsewhere. Then*

$$q = \frac{(1-s)^2}{2}, \qquad \alpha_1 = \frac{1-s}{2}, \qquad \alpha_1 - q = \frac{s(1-s)}{2}, \qquad p - \alpha_1 = \frac{s(3-s)}{2},$$

and

$$\|g\|_2^2 = sq^2 + (1-s)(\alpha_1 - q)^2.$$

Consequently

$$\frac{\|g\|_2^2 (p-\alpha_1)^2}{p\alpha_1(\alpha_1 - q)^2} = 9s + O(s^2) \to 0.$$

Thus no universal positive constant can make the rank-one lower bound

$$\|g\|_2^2 \gtrsim \frac{p\alpha_1(\alpha_1 - q)^2}{(p-\alpha_1)^2}$$

valid in the one-frontier regime.

Proof. The eigenfunction distinguishing the two cliques is mean-zero and has eigenvalue $(1-s)/2$. The degree is $(1-s)/2$ on the two cliques and 0 on the remaining set of measure s, which gives the formula for q and $\|g\|_2^2$. The displayed ratio is then a direct expansion at $s = 0$. □

Remark 14.3 (Current global status)**.** *The high-density theorem closes Region I. The remaining part of Conjecture 1.1 is therefore the one-frontier range* $1/3 < q < 1/2$*, equivalently* $1/2 < p < 2/3$*. In that range one needs a payment lemma converting the aggregate forced coupling in Lemma 14.1 into nonlinear spectral shift. The two-clique family in Proposition 14.2 shows that the payment cannot be just the rank-one forced-coupling bound with* b_1^2 *replaced by* $\|g\|_2^2$*.*

15 Accompanying files and certificate status

This package contains the following files.

- `odd_cycle_high_density_baton.tex`: this note.
- `odd_cycle_high_density_baton.pdf`: compiled version.
- `clean_two_sided_shift_checker.py`: exact rational checker for the two-sided identity, the path-shift expansion, the mixture identity, the kernel form, the finite strip certificate through $m = 61$, and several Region-II diagnostics.
- `clean_checker_m61.log`: successful run log for the finite strip certificate.
- `regionI_constant_certifier_exact.py`: exact rational certificate for the constants used in Appendix A.
- `regionI_constant_certifier_exact.log`: successful run log for the constant certificate.

The high-density theorem is therefore a small computer-assisted proof only in the finite range $m \leq 61$ and in the rational constant inequalities in Appendix A; the infinite tail $m \geq 63$ is otherwise analytic.

A Constant inequalities for the residual-strip closure

This appendix justifies the two numerical inequalities used in Lemma 11.2. Let

$$P(\theta) = \frac{2/3 - 2\theta}{\theta(1/2 - 2\theta)^2},$$

$$B_0(\theta) = \left(\frac{7}{6}\right)^{1-2\theta} (8 - 24\theta)^{-\theta} \frac{1/2 + \theta}{5/12 + \theta},$$

and

$$B_1(\theta) = \left(\frac{7}{6}\right)^{1-2\theta} (8 - 24\theta)^{-\theta} \frac{34}{29}.$$

The required inequalities are

$$\frac{99}{100m} P(\theta) B_0(\theta)^m \geq 1 \qquad (\theta = r/m \leq 1/6), \tag{36}$$

$$\frac{99}{100m} P(\theta) B_1(\theta)^m \geq 1 \qquad (1/6 \leq \theta = r/m < 1/4), \tag{37}$$

for the integer pairs appearing in the residual strip with $m \geq 63$ and $r \geq 2$.

For (36), the exact script first checks every integer pair with odd $63 \leq m \leq 499$. For the tail $m \geq 500$, use the elementary bounds

$$P(\theta) \geq 51, \qquad B_0(\theta) \geq \frac{201}{200} \qquad (0 < \theta \leq 1/6).$$

The first is the polynomial inequality

$$\frac{2}{3} - 2\theta - 51\theta \left(\frac{1}{2} - 2\theta\right)^2 \geq 0 \qquad (0 < \theta \leq 1/6).$$

For the second, $\log B_0$ is decreasing on $[0, 1/6]$. One way to see this is to set $t = 8 - 24\theta \in [4, 8]$ and use

$$\log t \geq \frac{2(t-1)}{t+1}.$$

The remaining rational inequality is

$$\frac{2(t-1)}{t+1} - \frac{8-t}{t} + \frac{48}{(20-t)(18-t)} = \frac{3t^4 - 123t^3 + 1462t^2 - 2888t - 2880}{t(t-20)(t-18)(t+1)} > 0$$

on $[4, 8]$, which is elementary. Hence

$$B_0(\theta) \geq B_0(1/6) = \frac{4}{\sqrt[3]{63}} > \frac{201}{200}.$$

Since $(201/200)^m/m$ is increasing for $m \geq 200$, it remains only to check

$$\frac{99}{100} \frac{51}{500} \left(\frac{201}{200}\right)^{500} > 1,$$

which is an exact rational inequality recorded by the constant certifier.

For (37), the exact script again checks all odd $63 \le m \le 499$. For the tail and, in fact, for all $m \ge 63$, use

$$P(\theta) \ge 72 \qquad (1/6 \le \theta < 1/4),$$

which follows from

$$\frac{2}{3} - 2\theta - 72\theta\left(\frac{1}{2} - 2\theta\right)^2 = -\frac{2}{3}(6\theta - 1)(72\theta^2 - 24\theta + 1) \ge 0$$

on $[1/6, 1/4]$. The uniform bound

$$B_1(\theta) \ge \frac{126}{125} \qquad (1/6 \le \theta \le 1/4)$$

is proved by splitting $[1/6, 1/4]$ into the twelve intervals

$$\left[\frac{1}{6} + \frac{i}{144}, \frac{1}{6} + \frac{i+1}{144}\right], \qquad 0 \le i \le 11.$$

On an interval $[a, b]$, the lower bound

$$B_1(\theta) \ge \left(\frac{7}{6}\right)^{1-2b} (8 - 24a)^{-b} \frac{34}{29}$$

holds. Since a, b are rational, the inequality that this lower bound is at least $126/125$ becomes an exact integer comparison after raising to a common denominator. The file `regionI_constant_certifier_exact.py` performs these twelve integer comparisons. Finally, $(126/125)^m/m$ has its minimum for $m \ge 63$ at $m = 125$ or 126, and the exact inequality

$$\frac{99}{100}\frac{72}{125}\left(\frac{126}{125}\right)^{125} > 1$$

completes the proof of (37).